

Technology Stack

Date	23 June 2026
Team ID	SWTID-2026-1377
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML5, CSS3, JavaScript, Flask (Jinja2 Templates)	Used to create a responsive and interactive user interface. JavaScript integrates the browser's Speech Recognition API for voice input, while Flask Jinja2 templates dynamically render web pages.

2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python 3, Flask	Flask serves as the backend framework, handling HTTP requests, processing user responses, coordinating NLP analysis, and generating personalized feedback and mastery scores
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	JSON Files (Topic Knowledge Base), Python Dictionaries, Temporary Session Storage	The project stores reference concepts and expected answers in JSON files. Since it evaluates responses in real time and does not require user account management, a traditional database is not necessary.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Local Flask Development Server (Can be deployed on Render, Railway, AWS, or Heroku in the future)	The application is currently designed for local development and testing. Flask's lightweight architecture makes it easy to deploy to cloud platforms when required.
5	Version Control & CI/CD	Git, GitHub	Git is used for version control, while GitHub hosts the project repository, enabling source code management, tracking changes, and collaboration.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Web Speech API, Scikit-learn, NumPy, NLTK, Sentence Transformers (optional), Pickle</i>	<i>Web Speech API captures voice input from the browser. Scikit-learn performs TF-IDF vectorization and cosine similarity calculations. NumPy supports numerical operations. NLTK is used for text preprocessing, while Sentence Transformers enhance semantic similarity analysis when available. Pickle is used for storing reusable NLP resources and configurations.</i>